E. Edge Queries

time limit per test: 1 second
memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

You are given a **directed unweighted** graph with ***N*** nodes and ***M*** edges. The nodes are numbered from 1 to N. Your task is to find the difference of indegree and outdegree of each node in the graph.

Input

The first line contains two integers N and M $(1 \le N \le 2 \times 10^5, 0 \le M \le 3 \times 10^5)$ — the number of vertices and the total number of edges.

The second line contains M integers $u_1, u_2, u_3 \ldots u_m$ $(1 \le u_i \le N)$ — where the i-th integer represents the node that is one endpoint of the i-th edge.

The third line contains M integers $v_1, v_2, v_3 \ldots v_m$ $(1 \le v_i \le N)$ — where the i-th integer represents the node that is other endpoint of the i-th edge.

The i-th edge of this graph is from the i-th node in the second line to the i-th node in the third line.

Output

Output a single line with N space-separated integers, where the i-th integer is the difference of indegree and outdegree of node i.

Examples

input Copy
```
5 10
2 5 4 3 2 4 3 4 1 3
5 1 5 5 1 2 2 1 3 4
```
output Copy
```
2 0 -2 -2 2
```

input Copy
```
5 4
5 3 3 2
1 1 2 4
```
output Copy
```
2 0 -2 1 -1
```

input Copy
```
8 7
7 7 7 2 1 4 1
2 6 3 4 2 8 5
```
output Copy
```
-2 1 1 0 1 1 -3 1
```

CSE221 Assignment 04 Summer 2026

Contest is running

17:25:12

Contestant

→ Submit?

Language: Java 21 64bit

Choose file: Choose File No file chosen

Success!

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